FORTISAID LAB 5

Securing Azure Storage with Virtual Network Service Endpoints

Environment Name: FortisAid
Region: West US 2
Resource Group: `fortisaid-lab05`

Objective

Implement network-level protection for Azure Storage by:

- Restricting storage access to selected virtual networks
- Enabling Service Endpoints
- Validating restricted access from Public VM
- Validating successful access from Private VM
- Mounting Azure File Share securely over SMB (Port 445)

This lab demonstrates enterprise-grade storage network isolation.

Architecture Overview

- **VNet:** `fortisaid-vnet05` (10.0.0.0/16)
- **Subnets:**
  - `Public`
  - `Private`
- **VMs:**
  - `FortisAid-VM-Public05`
  - `FortisAid-VM-Private05`
- **Storage Account:** `fortisaidstorage05`
- **File Share:** `fortisaid-files`

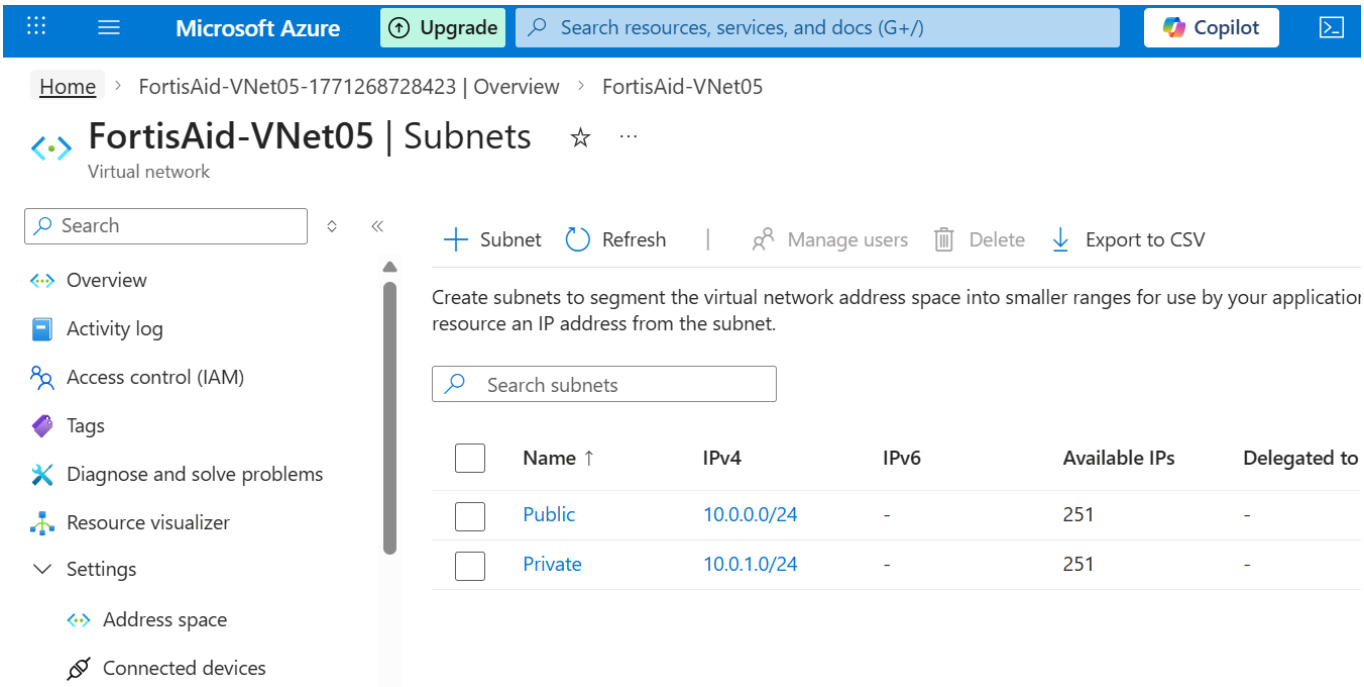
Exercise 1 – Deploy Core Infrastructure

Step 1 – Create Resource Group

```
New-AzResourceGroup -Name fortisaid-lab05 -Location "West US"
```

Step 2 – Create Virtual Network and Subnets

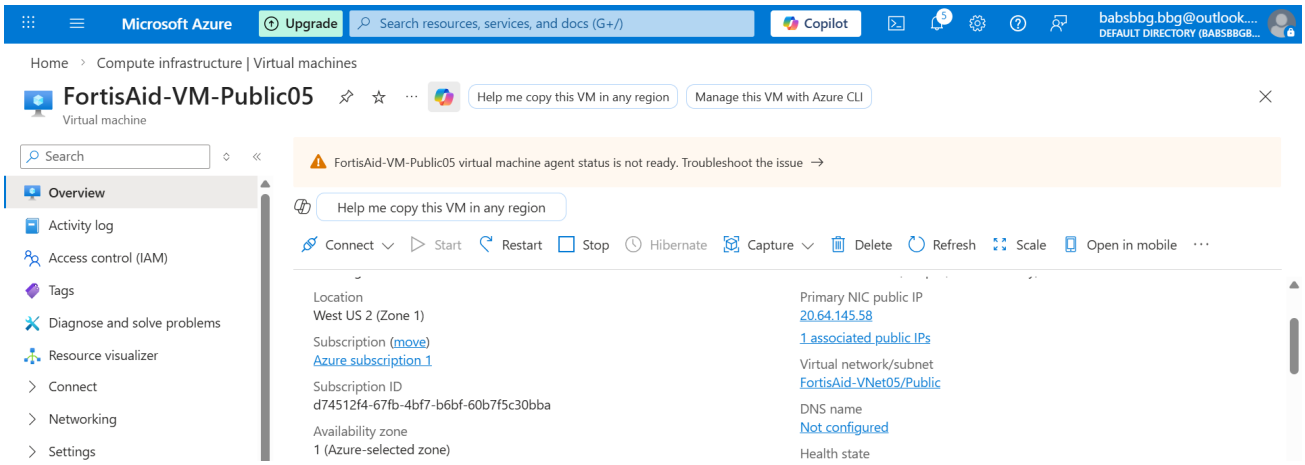
- Address Space: 10.0.0.0/16
- Public Subnet
- Private Subnet

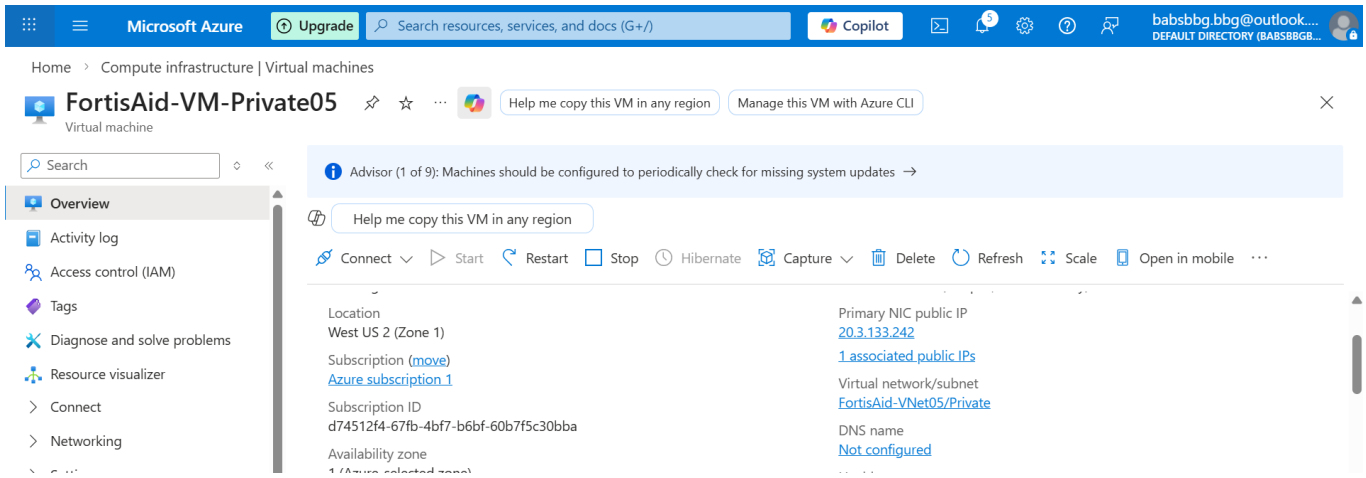


VNet with Public and Private Subnets

Step 3 – Deploy Virtual Machines

- Deploy one VM into Public subnet
- Deploy one VM into Private subnet
- Enable RDP for both

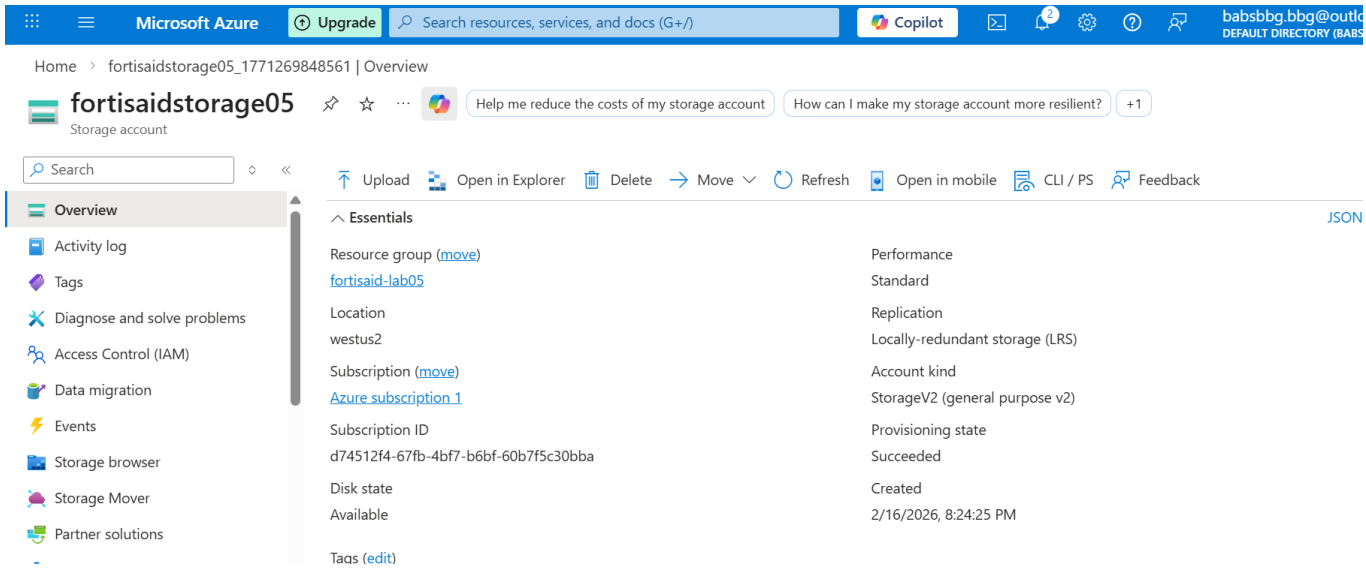




Both VMs Running`

Step 4 – Create Storage Account

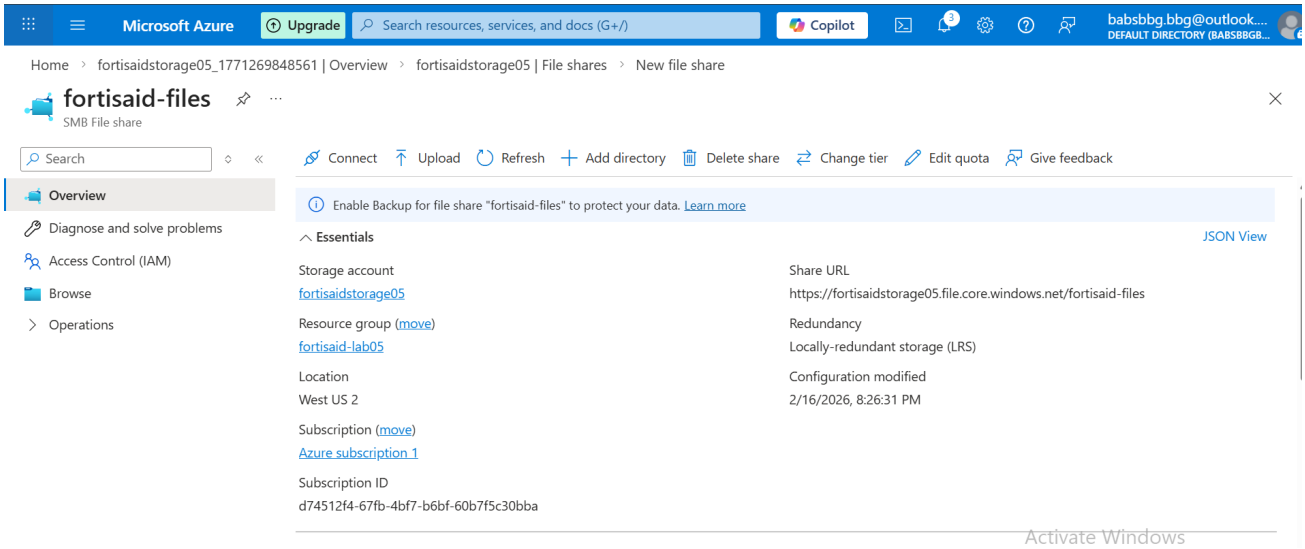
- Name: `fortisaidstorage05`
- Performance: Standard
- Redundancy: LRS
- Secure transfer required: Enabled



Storage Account Overview`

Step 5 – Create Azure File Share

- Name: `fortisaid-files`

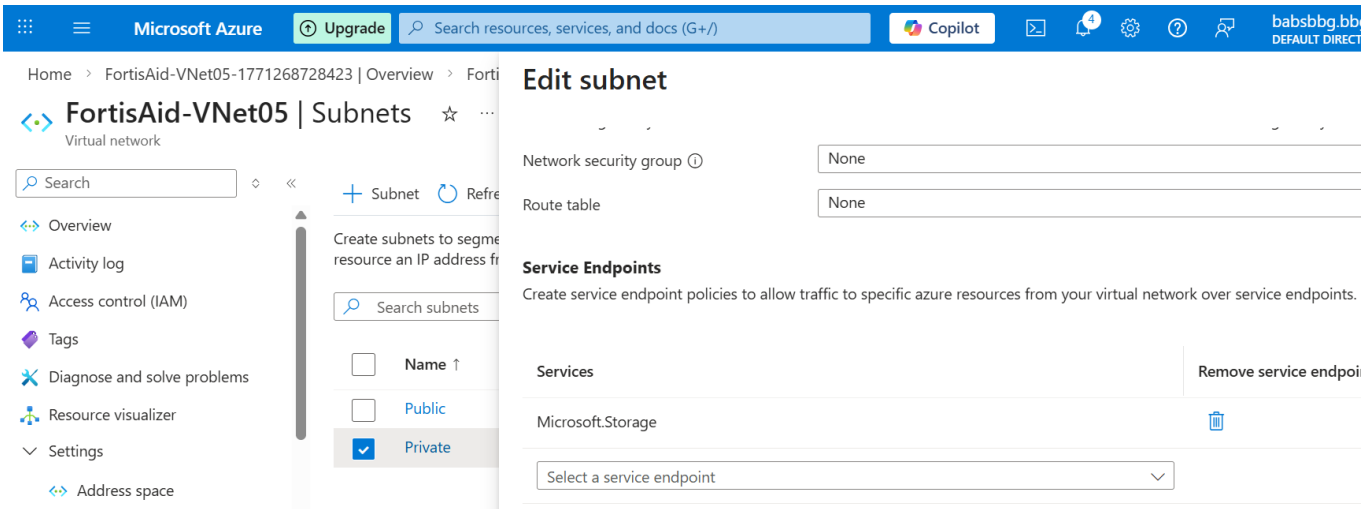


File Share Created`

Exercise 2 – Enable Service Endpoints

Step 6 – Enable Service Endpoint on Private Subnet

1. Go to VNet
2. Select **Private subnet**
3. Enable service endpoint for:
    - Microsoft.Storage



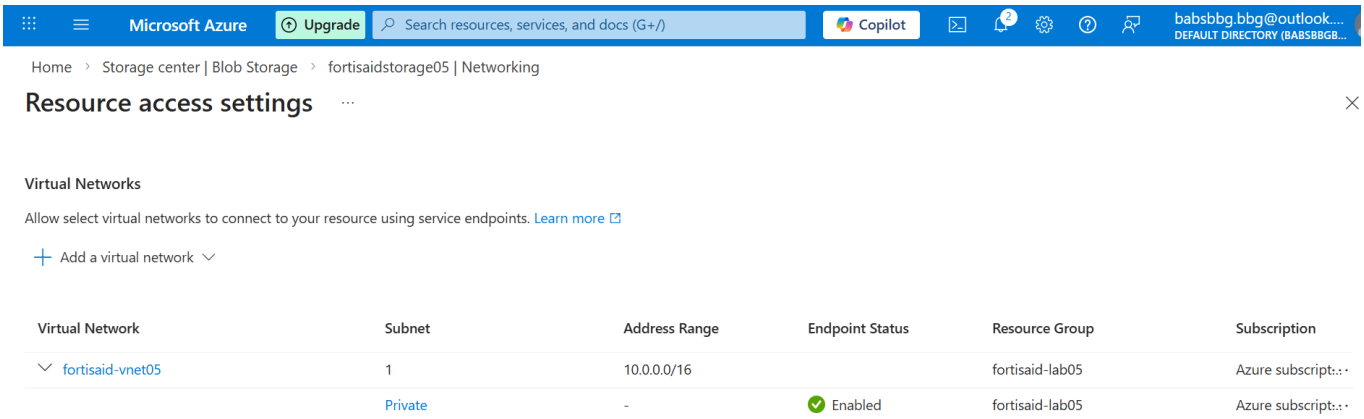
Service Endpoint Enabled on Private Subnet`

Step 7 – Restrict Storage to Selected Networks

1. Go to Storage Account → Networking
2. Select:
    - **Selected networks**

3. Add:
    - `fortisaid-vnet05`
    - Subnet: `Private`

Save changes.



Storage Networking Restricted to Private Subnet

Exercise 3 – Validate Network Isolation

Step 8 – Test from Public VM (Expected Failure)

RDP into:

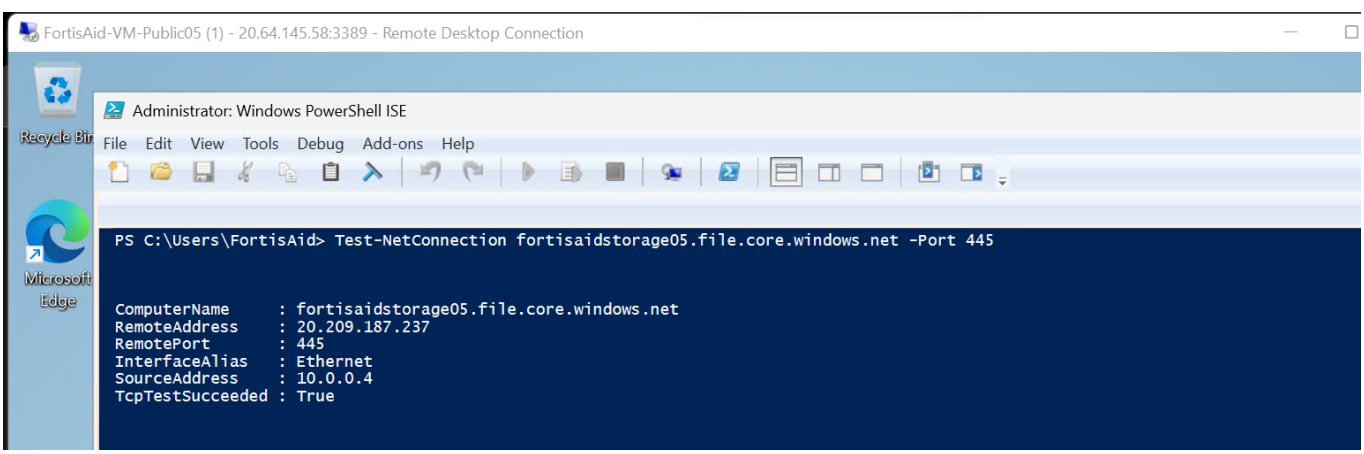
`FortisAid-VM-Public05`

Test Port 445

`Test-NetConnection fortisaidstorage05.file.core.windows.net -Port 445`

Result:

`TcpTestSucceeded : True`



Public VM Port 445 Test`

Attempt to Mount Share (Expected Failure)

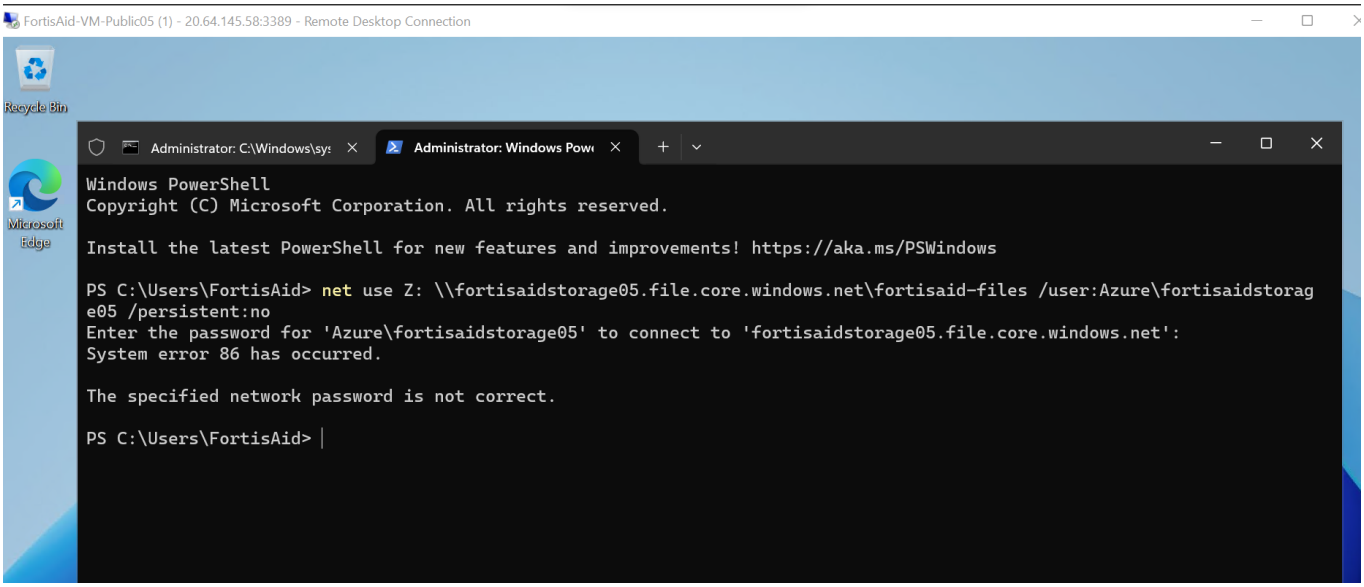
```
net use Z: \\fortisaidstorage05.file.core.windows.net\fortisaid-files
/user:Azure\fortisaidstorage05
```

Enter storage key when prompted.

Expected result:

```
System error 86 has occurred. The specified network password is not correct.
```

Although SMB port is reachable, authentication fails because Public subnet is not authorized.



Public VM Mount Fails with Error 86`

Step 9 – Test from Private VM (Expected Success)

RDP into:

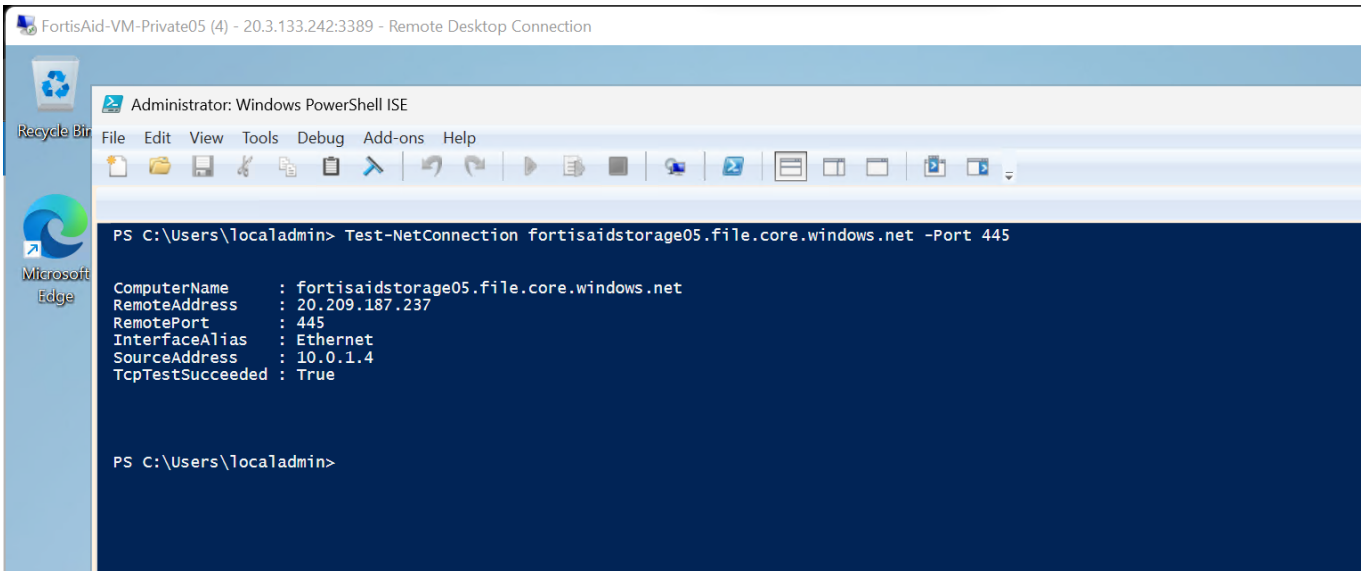
```
FortisAid-VM-Private05
```

Confirm Port 445 Reachable

```
Test-NetConnection fortisaidstorage05.file.core.windows.net -Port 445
```

Expected:

```
TcpTestSucceeded : True
```



Private VM Port 445 Test`

Mount Azure File Share

`net use Z: \\fortisaidstorage05.file.core.windows.net\fortisaid-files /user:Azure\fortisaidstorage05`

Enter storage key when prompted.

Expected:

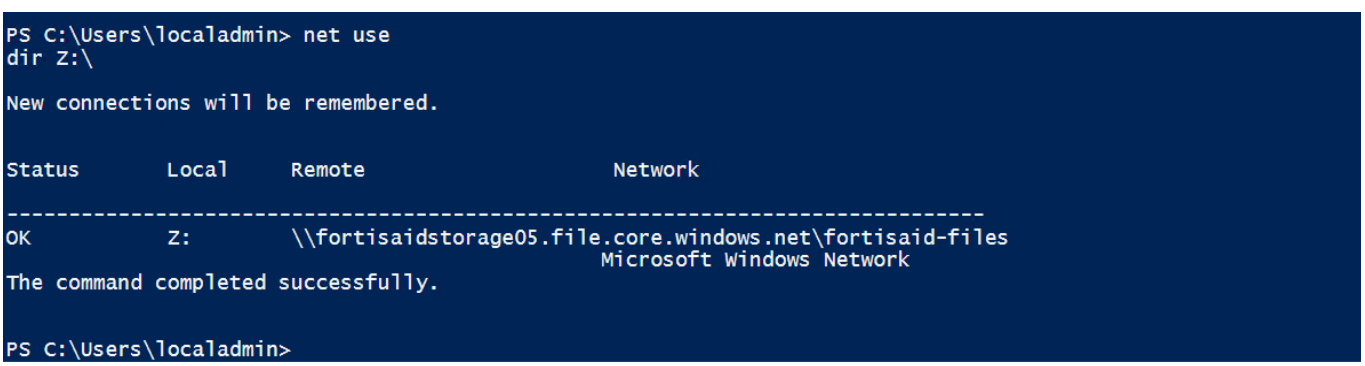
`The command completed successfully.`

Verify:

`dir Z:\`

Expected:

- Directory listing visible



Private VM Successful Mount of Z Drive`

Security Validation Summary

VM	Port 445	SMB Mount	Expected
Public VM	True	❌ Fails (Error 86)	Correct
Private VM	True	✅ Success	Correct

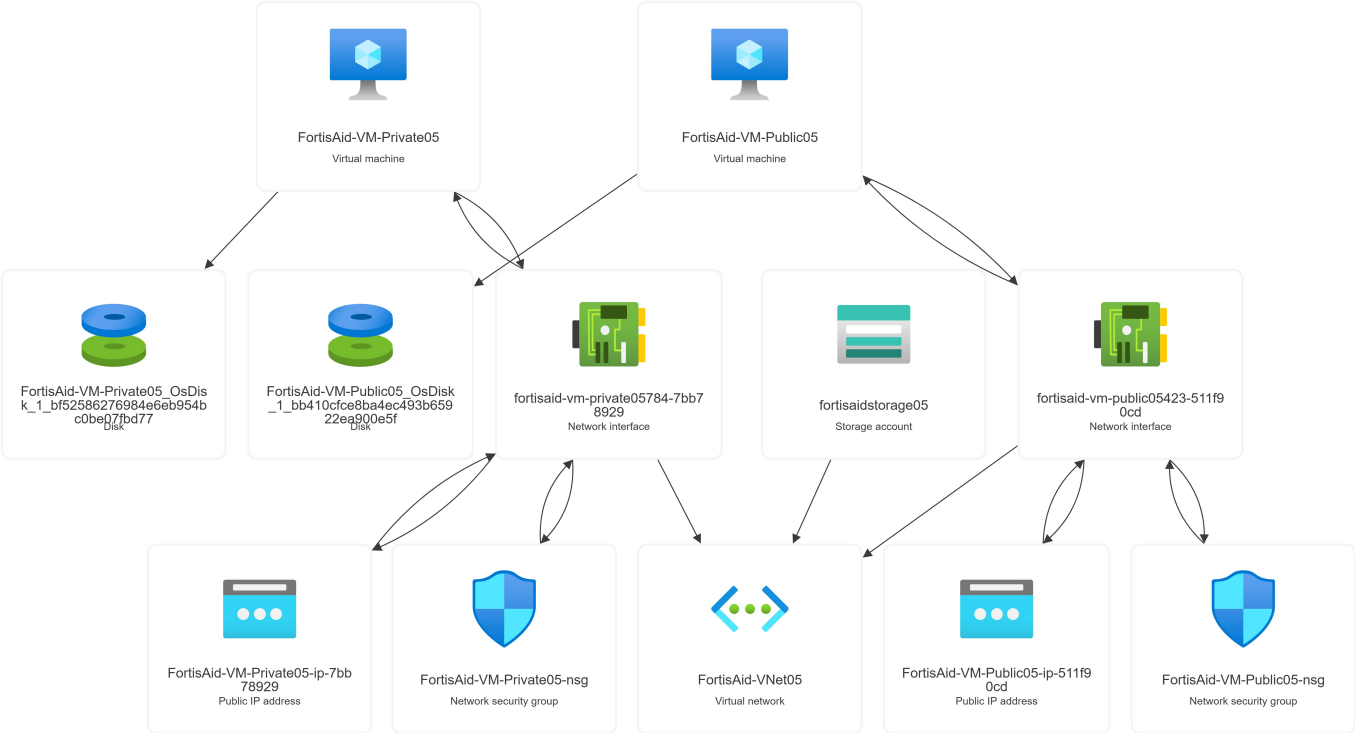
This proves:

- Storage is reachable at network level
- Access is restricted at subnet authorization level
- Only approved subnet can authenticate and mount

What I Learned as a Cloud Security Engineer

1. **Port Reachability ≠ Authorization**
   - TCP 445 being reachable does not mean access is permitted.
2. **Service Endpoints enforce subnet identity**
   - Storage validates the source subnet before allowing authentication.
3. **Azure Storage network rules override valid credentials**
   - Even correct keys fail if subnet is not allowed.
4. **Enterprise network hardening requires layered validation**
   - Port testing
   - Network rules
   - Authentication testing

Successfully implemented:

- Azure Storage network isolation
- Service Endpoint configuration
- Subnet-based access control
- SMB validation testing

ARCHITECTURAL DIAGRAM